* find out the list of companies which are conducting the placement ~~rights~~ drive using DSA course

→ findout the syllabus of gate for DSA.

→ find out set of I & O for the following.

① Electricity bill

② Bus ticket

③ Message to a friend

→ today when you started from ~~home~~ hostel to college what are the various data structures you identified in your surroundings.

gate syllabus

1. Programming & Data Structures
   - → C programming basics
   - → Recursion
   - → Arrays
   - → Stacks
   - → Queue
   - → Linked Lists
   - → Trees
   - → Binary Search Trees
   - → Binary Heaps
   - → Graphs
2. Algorithms
   - → searching - Linear search
     - Binary search
   - → sorting - Bubble sort, selection sort, Insertion sort, Merge sort, Quick sort, Heap sort
   - → Hashing
   - → Time and space complexity - Big-O(O)
     - Omega(Ω)
     - Theta(θ)
   - → Algorithm Design Techniques - Divide and conquer
     - Greedy Algorithms
     - Dynamic programming
   - → Graph Algorithms - BFS
     - DFS
     - Minimum spanning tree
     - Shortest path.

Tier 1 (very high DSA)

- Google
- Microsoft
- Amazon
- Meta
- Apple
- NVIDIA
- Uber
- Atlassian
- Adobe
- Salesforce

Tier 2 (strong DSA)

- Flipkart
- Walmart
- Intuit
- Oracle
- Paypal
- Visa
- American Express
- Goldman Sachs
- Morgan Stanley
- D.E.Shaw

Indian product companies

- Phonepe
- Razorpay
- Meesho
- Groww
- CRED
- Freshworks
- Zoho
- Juspay
- Swiggy
- Zomato

Service Based companies (Basic to moderate DSA)

- TCS
- Infosys
- Wipro
- Cognizant
- Accenture
- Capgemini
- HCLTech
- Tech Mahindra
- IBM

③ Electricity bill:

Input:

- Consumer Number
- Customer Name
- Previous Meter Reading
- Current Meter Reading
- Rate per unit
- Fixed charge

Output:

- Units Consumed
- Electricity Bill Amount
- Due date
- payment status

2) Bus ticketing

Input:	Output:
→ passenger Name	→ Ticket Number
→ source	→ Passenger Name
→ Destination	→ Seat Number
→ Date	→ Fare
→ Bus Type	→ Bus Number
→ Number of Tickets.	→ Journey Date

3) Sending a message to friend.

Input	Output:
→ Sender Name	→ Receiver
→ Receiver Name	→ Message
→ Message	→ Date & Time
→ Date	→ status.
→ Time	

4) Mess token — object

- plates — array
- classroom benches — array
- Books — array
- Subjects — array
- attendence register — array

→ Create a structure variable student and estimate the size of the structure manually then write a C-program to find size of a structure using size of operator now observe both the outputs. Are they same or different? Justify your answer.

→

```
#include<stdio.h>
struct student
{
   int rollno;
   char grade;
   float marks;
};
int main()
{
   struct student s;
   printf("size of structure = %zu bytes\n", sizeof(s));
   return 0;
}
```

Manual

int – 4
char – 1
float – 4
} total = 9bytes

Program o/p:

Size of structure = 12bytes

No, they are different. Because of structure padding (alignment). The compiler inserts extra unused bytes b/w members or at the end of the structure so that data is aligned in memory for faster access by the CPU.

→ ① Open amazon website and find out 10 different functions.

→ ② Return 1 or return 0 means what, the main returns to whom.

①A: (1) Search products (2) Product categories
(3) Add to cart (4) Buy Now (5) Wishlist
(6) Customer Reviews & Ratings (7) Order Tracking
(8) Filter and sort (9) User login (10) Payment

②(A) It return the value to operating system (os).
return 0: program executed successfully.
return 1: program did not complete successfully.
It contains general error / failure.

→ Algorithm

Q: In java every object has hash code findout what is hash code and why it is being generated in java.

what is hash code? - It is an integer value that represents an object. It is mainly used to identify objects efficiently in hash-based collections such as: HashMap, HashSet, HashTable

Why it is generated? to improve searching speed, helps compare objects efficiently, determine the bucket.

eg:

```
class Demo{
    public static void main(String[] args){
        String s = "Hello";
        System.out.println(s.hashCode());
    }
}
```

o/p:

6960650.

→ HW1

compute that time complexity using frequency count for the following.

① `for(i=1; i<=10; i=i+2)`	1+1+2	1+11+20	32
`{ i = i&10;`	1+1	20	20
`printf("%d", i);`	1	10	10
`}`			62
② `i = 1;`	1	1	1
`while (i<=50);`	1		1
`{ printf("hello");`			2
`}`			
③ `for(i=1; i<=20; i++);`	1+1+1	3	3
④ `for(i=1; i<=n; i=i++)`	1+1+2	1+(n+1)+2n	3n+2
`{`			
`printf("hello");`	1	n	n
`printf("world");`	1	n	n
`printf("i=%d", i);`	1	n	n
`}`			6n+2